

Ligjerata 11ë

Puna e gazit të mbyllur gjatë procesit izohorik

$$V = \text{konst.}$$

$$dV = 0$$

$$A = \int p dV = 0$$

$$A = 0$$

$$T \uparrow, U \uparrow$$

$$dQ = dU + p dV \quad / : dT$$

$$\frac{dQ}{dT} = \frac{dU}{dT} + p \frac{dV}{dT}$$

$$V = \text{konst.}$$

$$dV = 0$$

$$C_V = \frac{dU}{dT}$$

$$dU = C_V dT$$

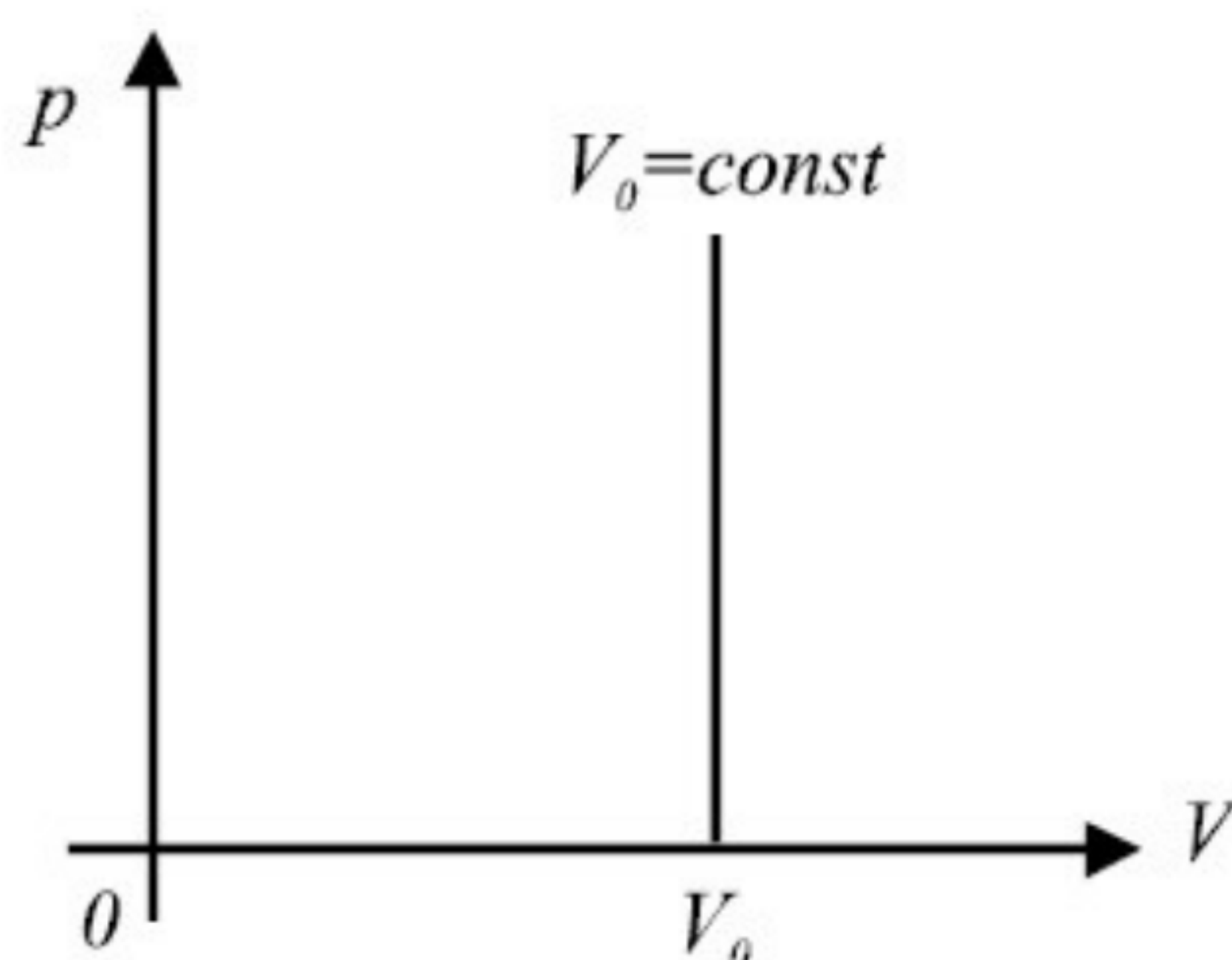
$$V = \text{konst.}$$

$$\Delta V = 0$$

$$Q = \Delta U = U_2 - U_1$$

$$Q = n C_V \Delta T$$

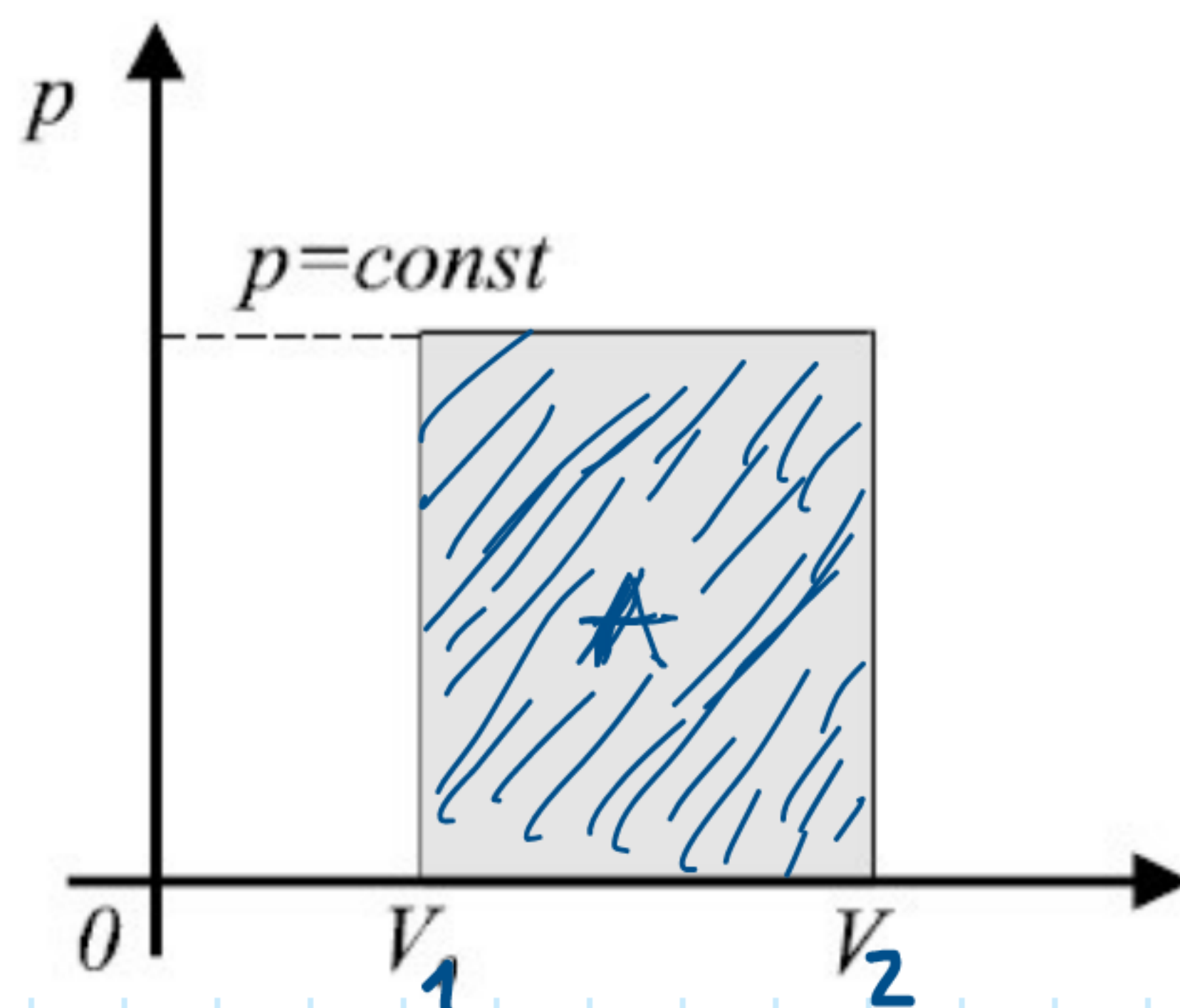
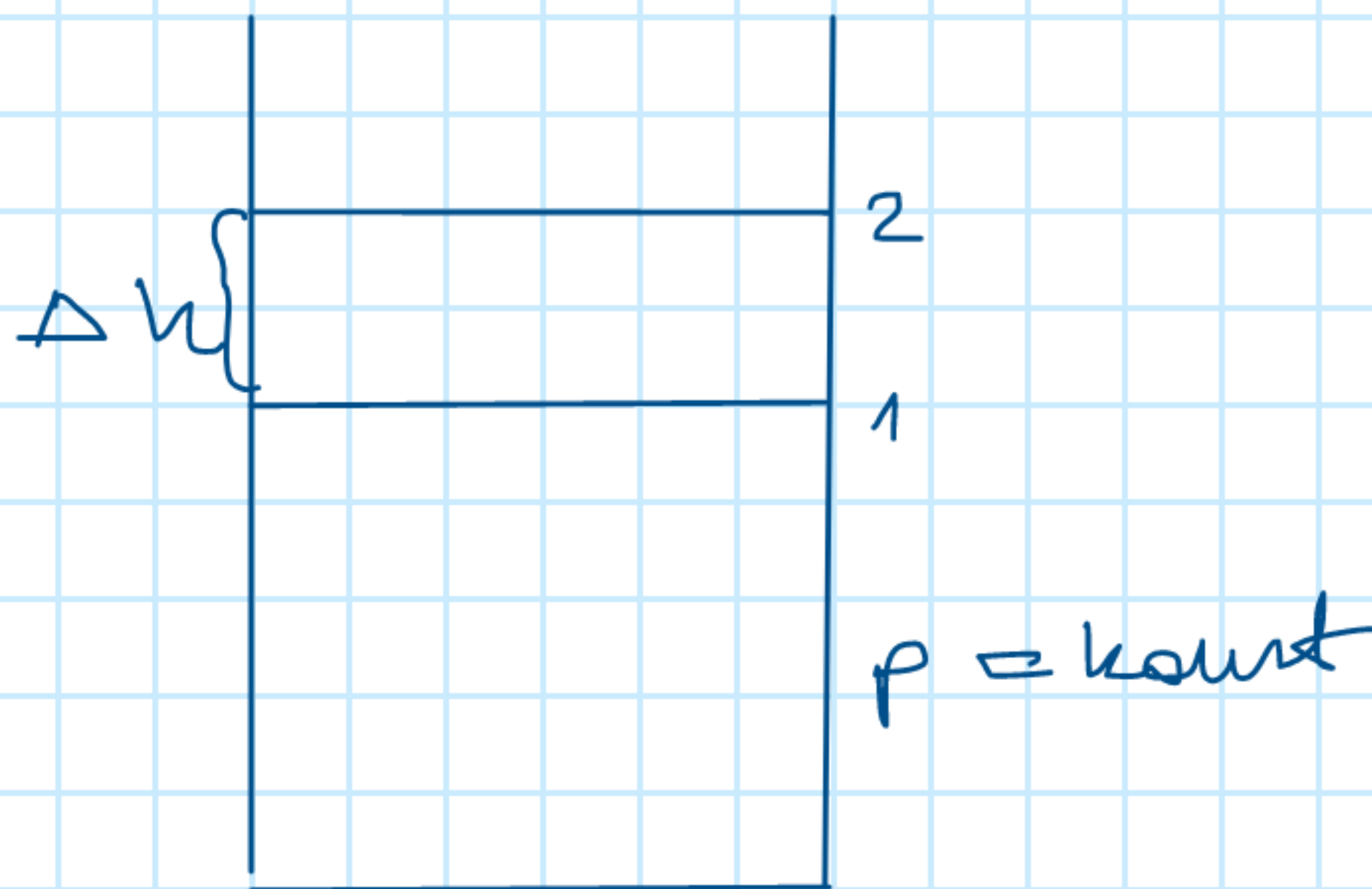
$$\Delta U = n C_V \Delta T$$



Puna e gazit të mbyllur gjatë procesit izobarik

Gaz i ngrohet $T \uparrow$ sepse $p = \text{konst.}$, duhet të rritet $V \uparrow$

$$A = \int F dh$$



$$A = \int_1^2 F dh = \int_1^2 p s dh = p s (h_2 - h_1) = p (V_2 - V_1) = p \Delta V$$

$$\Delta Q = \Delta U + \Delta A$$

$$\Delta Q = \Delta U + p \Delta V$$

$$U = \frac{i}{2} n R T$$

$$p \Delta V = n R \Delta T$$

$$\Delta U = \frac{i}{2} n R \Delta T$$

$$\Delta Q = \frac{i}{2} n R \Delta T + n R \Delta T$$

$$\Delta Q = \frac{i+2}{2} n R \Delta T$$

$$p \cdot \Delta V = n R T$$

$$U = \frac{i}{2} p \cdot \Delta V = \frac{i}{2} p (V_2 - V_1)$$

$$Q = \frac{i+2}{2} p (V_2 - V_1)$$

$i \rightarrow$ numuri i shkallëve të lirisë të molekulave të gazit.

Puna e gazit të mbyllur gjatë procesit izotermik

$$T = \text{konst.}, \quad U = \text{konst.}, \quad V_1 \rightarrow V_2$$

$$U_1 = U_2 \rightarrow dT = 0$$

$$\Delta U = U_2 - U_1 = 0$$

$$Q = A$$

$$A = \int_{V_1}^{V_2} p dV, \quad p \cdot V = n R T$$

$$p = \frac{n R T}{V}$$

$$A = \int_{V_1}^{V_2} n R T \frac{dV}{V} = n R T \int_{V_1}^{V_2} \frac{dV}{V} = n R T \left(\ln V \right) \Big|_{V_1}^{V_2}$$

$$A = n R T \ln \frac{V_2}{V_1}$$

$$p_1 V_1 = p_2 V_2 \Rightarrow \frac{p_1}{p_2} = \frac{V_2}{V_1}$$

$$A = n R T \ln \frac{p_1}{p_2}$$

